

Plan for Collecting and Using Data

1 Objective of Data Collection

To gather high-quality, diverse textual data that helps the machine learning model accurately detect fake news in real time across various domains such as politics, health, finance, entertainment, and social issues.

2 Data Sources

Public Datasets

- LIAR Dataset - statements labeled as true/false.
- FakeNewsNet Dataset - Includes BuzzFeed, PolitiFact fake/real news.
- Kaggle Fake News Dataset - Contains labeled news articles for binary classification.
- ISOT Fake News Dataset - Real vs fake news articles dataset.

Live Data Sources

- News Websites - APIs such as NewsAPI.org.
- Social Media - Twitter/X streaming API.
- RSS Feeds - Various news channels provide RSS feeds for live updates.

3 Data Collection Method

Automated Data Scraping & API Fetching

- Python scripts for scraping news content from websites.
- API integration using News API, Twitter API, Google News RSS.

Manual Data Verification

- Team members manually label a portion of collected stories.
- Cross-check with trusted fact-checking sources like PIB, PolitiFact, Reuters.

4 Data Preprocessing Steps

Cleaning

- Remove stop words
- Remove HTML tags
- Handle special characters
- Normalize text

Tokenization & Vectorization

- TF-IDF
- BERT Sentence Embeddings

Handling Imbalanced Data

- Oversampling fake news examples
- SMOTE technique

5 Data Usage Plan

Training the ML Model

- 70% =Training
- 15% =Validation
- 15% =Testing

Real-Time Prediction

Live data will be fetched and processed to provide fake probability score and classification.

3 Continuous Model Improvement

Newly verified news items will be stored and model retrained periodically.

6 Data Storage & Security

- Data stored in MySQL.
- Processed datasets stored in CSV.

Security Measures

- API keys stored securely.
- No private user data collected.
- Only public data used.

7 Ethical & Legal Considerations

- Follow API and platform terms.
- Avoid collecting personal data.
- Use data for educational/research purpose only.

8 Tools & Technologies

- Python
- TensorFlow
- MySQL
- NewsAPI, Twitter API, RSS Readers